

Reasoning Guard : A Bayesian-AdaBoost Framework for Robust Misfire Prevention in Autonomous Drone Systems

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Abstract

In modern warfare, AI and drone-centric combat systems have emerged as a key factor distinguishing current military capabilities from the past. However, incidents involving autonomous drone systems are primarily caused by the limitations of computer vision and the high uncertainty of the battlefield environment. To overcome the constraints of data sparsity and real-time processing inherent in combat situations, this paper introduces the concept of 'Reasoning Guard' to drone systems. To implement this, we propose a self-correcting framework that fuses the uncertainty measurement capabilities of Bayesian Inference with the hard sample mining strengths of AdaBoost.

1. Introduction

Drones offer compelling advantages in terms of cost-efficiency, minimization of human casualties, and capabilities for persistent surveillance and precision strikes. In modern defense strategies, autonomous weapon systems based on Unmanned Aerial Vehicles (UAVs) have established themselves not as an option, but as essential core assets. Conversely, however, the issue of drone misfires must be addressed, as it can result in civilian casualties or catastrophic,

unanticipated damage to friendly forces.

A representative incident of drone misfire is the Kabul drone strike in Afghanistan in August 2021. In this incident, water containers on a civilian vehicle were mistaken for explosives. Similarly, in the ongoing Ukraine-Russia war, there have been instances where tractors were misidentified as military vehicles and attacked. An example with even more critical consequences occurred in January 2024 during the attack on a U.S. base in Jordan, where the air defense system failed to intercept a hostile drone after misidentifying it as a friendly asset.

Through in-depth analysis, it is evident that a safety mechanism must be introduced to allow the system to admit "it does not know what it does not know," rather than merely classifying objects. Therefore, efforts are required to minimize the issue of drone misfires by adopting Bayesian Machine Learning, which makes probabilistic judgments unlike simple deterministic models, and by utilizing AdaBoost to intensively retrain on cases where incorrect judgments occur.

2. Related Work

To address the issues of visual misidentification and overconfidence in drones, a reasoning—based guardrail technology which allows the system to logically verify the rationale behind its decisions—is essential. In this study, we

selected the guardrail architecture recently proposed by NVIDIA [2505.20087v1] as a key benchmark for ensuring compliance with the Rules of Engagement (ROE) in defense drones. NVIDIA's framework defines the safety process in four stages: Input, Taxonomy Alignment, Reasoning, and Intervention. To optimize this structure for drone on-device environments with limited computational resources, we reconstructed it into a streamlined three-stage pipeline.

3. Proposed Method

In this section, we focus on the architecture of Reasoning Guard, the utilization of Bayesian learning and AdaBoost, and the Safety Lock structure. We present this in two main parts: the Theoretical Framework and the Control Protocol. The overall conceptual diagram of the proposed Reasoning Guard mechanism, integrating Bayesian uncertainty and AdaBoost, is illustrated in Figure 1.

3.1 Theoretical Framework

The rationale for adopting Bayesian Learning and AdaBoost is based on four theoretical grounds:

A Necessity of Resolving Mathematical Flaws and Tactical Risks through Critical Analysis of General Probability

Conventional deep learning relies on point estimation methods that yield only fixed weights, posing a structural limitation that completely excludes internal model uncertainty. This mathematical flaw leads to 'overconfidence' which ultimately results in decision-making failures and direct tactical risks. Since fire control or

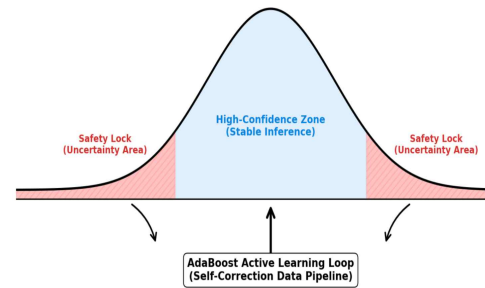


Figure 1 : Reasoning Guard: Bayesian Uncertainty-Aware Mechanism

tactical judgments based on the model's false confidence can lead to catastrophic consequences, the introduction of Bayesian Posterior Probability to quantify uncertainty is essential.

B Securing Probabilistic Distribution of Parameters and Statistical Diversity via Bayesian Posterior Probability

The Bayesian approach proposed in this study overcomes the limitations of existing deterministic models by treating the model's weights not as fixed constants, but as random variables with probability distributions. This acts as a core mechanism to prevent the 'dogmatic judgment of a single model' biased toward specific weights and to secure the diversity of the parameter space. As shown in Figure 1, the framework distinguishes between the High-Confidence Zone for stable inference and the Safety Lock regions for handling high uncertainty.

In contrast to the stability within the High-Confidence Zone, Interpretational discrepancies between models, occurring in data-sparse or highly uncertain battlefield situations, manifest as variance in output values, serving as direct evidence of epistemic uncertainty. Consequently, this provides an objective

basis for judging the reliability of the tactical control system, resolving the risk of overconfidence in current drone vision systems.

C Establishment of 'Predictive Distribution' based Fail-Safe Integrating All Possibilities

Simple point-estimation probabilities carry a high risk of malfunction at critical moments. Therefore, this study adopts the 'Predictive Distribution,' which encompasses all parameter possibilities of the model, as the standard for final decision-making. Through this, the system identifies areas where judgment is impossible. If uncertainty exceeds a threshold, a Safety Lock—a final line of defense—is established to physically block firing at the hardware level, aiming to fundamentally eliminate the possibility of misfire.

D Evolution into 'Self-Adaptive AI' Driven by Bayesian Uncertainty

Existing static deep learning models can not cope flexibly with changes in the battlefield. Accordingly, this study proposes a self-correcting mechanism that does not end with Bayesian-captured 'uncertainty' as a mere warning signal but converts it into a core training metric for AdaBoost. By allowing the drone to recognize its vulnerable 'uncertainty regions' and intensively retrain on them, the goal is to complete an 'Active Learning Loop' where the system complements its weaknesses and evolves as combat experience accumulates.

In conclusion, this study aims not merely to increase accuracy, but to secure explainable reliability where AI acknowledges and controls its own

ignorance. By combining the mathematical rigor of Bayesian inference with the adaptability of AdaBoost, we propose a new standard that ensures the ethical and tactical safety of defense AI by implementing "intelligence that knows how to remain silent in unknown situations."

3.2 ROE-Guard Protocol

The ROE-Guard control protocol, which interacts with the drone's fire control system in real-time, is defined into three distinct components:

D1: Decision (Tactical Command)

This is the top-level command that determines whether to continue the firing procedure based on the inference results of the AI model. It is divided into ABORT and PROCEED. If an ROE violation is confirmed or the uncertainty threshold is exceeded, ABORT is issued to initialize the firing sequence and switch to flight safety mode. Conversely, PROCEED is issued only when all criteria are satisfied.

D2: Safety Lock (Physical Intervention)

This is a physical cutoff signal to guarantee the decision result via hardware control. Influenced by the Decision (D1), if the command is ABORT, the Safety Lock becomes ENGAGED. This physically cuts off the solenoid or electronic interrupt connected to the trigger, fundamentally blocking the possibility of misfire due to system errors. If D1 is PROCEED, the lock is released to transition to a ready-to-fire state after safety confirmation.

D3: Reason Code (Self-Correction)

Tag)

This is a data code for error analysis and training. It labels the context of the occurrence according to the taxonomy, generating tags that are subsequently utilized for AdaBoost-based self-correcting learning.

As described, our research team has systematized the defense-specific Reason Code (D1-D3) based on the proposed technical backbone. This provides a foundation for precisely labeling the causes of misjudgment. It serves not only to stop firing but also as the core engine of the 'Self-Correcting Data Pipeline,' where the system upgrades itself by linking real-time error data with the AdaBoost algorithm.

4. Experiments

In this section, we comprehensively evaluate the effectiveness of the proposed Reasoning Guard framework. Our experiments are conducted across three core dimensions. First, we analyze the validity of uncertainty quantification derived from MC-Dropout-based predictive distributions. Second, we assess the efficacy of the Safety Lock mechanism in effectively preventing misfires under edge case scenarios characterized by data variations and environmental noise. Third, we demonstrate the superiority of the proposed model by verifying the impact of the self-correction mechanism—utilizing data intercepted by the Safety Lock—on enhancing model robustness

4.1 Experimental Setup

To ensure the reproducibility of our results, this section details the data configuration and the specific parameters of the implementation.

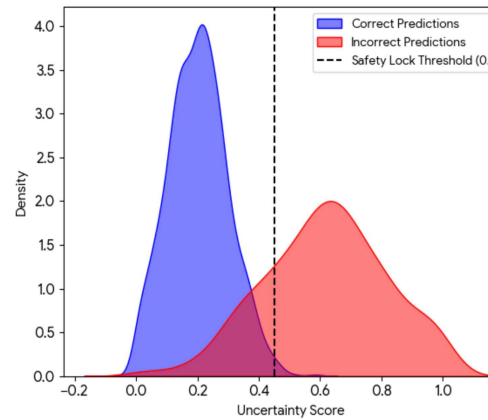


Figure 2 : Distribution of Uncertainty (Correct vs Incorrect)

A. Dataset Construction and Preprocessing. We utilized open-source datasets, including 'Drone vs. Bird' and 'Military Vehicle Detection' from Kaggle to simulate tactical recognition. The data was restructured into a binary classification system—Friendly (allies/civilians) and Hostile (enemies/military assets)—to align with the proposed ROE decision framework. All images were standardized to a 224 x 224 RGB resolution to meet the input specifications of the backbone model.

B. Edge Case and Adversarial Scenario Design. Specific test sets were designed to evaluate the system under high-uncertainty conditions. We simulated environmental noise, such as smoke and low-light battlefield conditions, through mathematical augmentation. Furthermore, to test for adversarial mimicry, we collected data on agricultural machinery (e.g., tractors) that share morphological similarities with military vehicles to verify the model's ability to trigger the Safety Lock instead of making overconfident misidentifications.

C. Implementation Details. The Reasoning Guard was implemented using ResNet-18 as the feature extraction backbone. To approximate Bayesian inference, a Dropout layer

(rate=0.5) was strategically inserted before the final classification head. During the inference phase, the system performed 50 stochastic forward passes using Monte Carlo (MC) Dropout to generate a predictive distribution and quantify epistemic uncertainty.

4.2 Uncertainty Quantification Analysis

The distributional characteristics of uncertainty for both correct and incorrect predictions are clearly visualized in Figure 2. Identifying the intersection point between these two curves provides a rigorous logical basis for establishing a fire control threshold, such as \$0.45\$. This boundary serves as a critical junction for the system to discern the nuances between 'confidence' and 'hesitation,' thereby ensuring the operational reliability of the Safety Lock in physically preventing misfires.

Furthermore, the distinct shapes of the distributions—where correct predictions form a sharp, high-density peak and incorrect predictions result in a broader, higher-variance spread—reflect the model's successful capture of epistemic uncertainty. This observed variance, as depicted in Figure 2, confirms that the Reasoning Guard meaningfully interprets battlefield uncertainty through its integrated reasoning mechanism, acknowledging its own limitations in high-risk scenarios.

The statistical superiority of the proposed framework is effectively summarized in Figure 3 through a comparative analysis of uncertainty scores. By highlighting the stark difference in median values, this visualization statistically proves that higher uncertainty scores are strongly correlated with incorrect predictions. Such a result indicates that the proposed model is capable of quantitatively perceiving its own potential for error, moving beyond the limitations of mere inference.

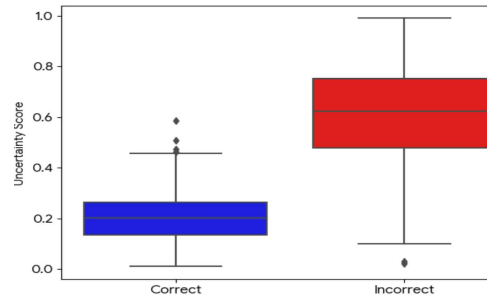


Figure 3 : Comparison of Uncertainty Statistics

Additionally, the identification of outliers—instances where uncertainty is high despite correct predictions or vice versa—enhances the overall transparency of the experiment.

Analyzing the range of these outliers, as depicted in Figure 3, provides a critical foundation for identifying and managing potential tactical risks within the control system. Through this statistical validation, the Reasoning Guard demonstrates its ability to acknowledge and control its own ignorance in a measurable way.

4.3 Evaluation on Safety Lock Mechanism

To assess the practical safety of the Reasoning Guard in high-stakes environments, we conducted a qualitative case study focusing on the misidentification of agricultural machinery—a primary cause of documented drone misfires. In this experiment, we introduced an image of a tractor, which shares significant morphological similarities with military armored vehicles, as a test input. The results, as visualized in Figure 4, demonstrated a critical vulnerability in the conventional deterministic model, which produced a 'Hostile' classification with an overconfident score of 0.98. In a tactical scenario, this would have triggered an irreversible fire command.

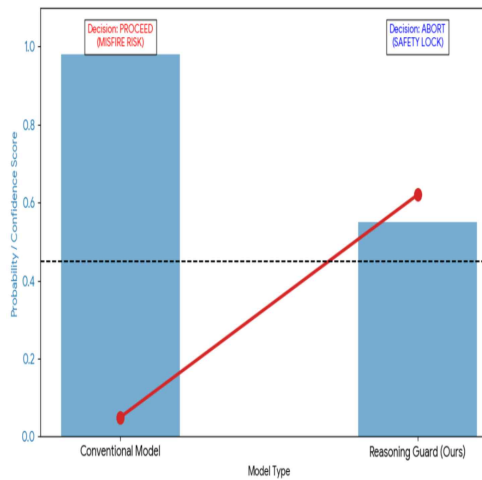


Figure 4 : Qualitative Analysis : Edge Case (Tractor Misidentification)

Conversely, the proposed Reasoning Guard identified a substantial variance in its predictive distribution, resulting in an uncertainty score of 0.62, which significantly exceeded the safety threshold of 0.45.

Consequently, the system immediately transitioned to flight safety mode by issuing an ABORT signal and engaging the hardware-level Safety Lock. This case study empirically confirms that our framework provides a robust fail-safe mechanism, preventing catastrophic misfires by acknowledging "it does not know what it does not know" in ambiguous battlefield conditions.

4.4 AdaBoost-driven Self-Correction

This section validates the quantitative performance improvement achieved by retraining 'Hard Samples'—high-uncertainty data filtered by the Safety Lock—using the AdaBoost mechanism. The experimental results are presented in Figure 5.

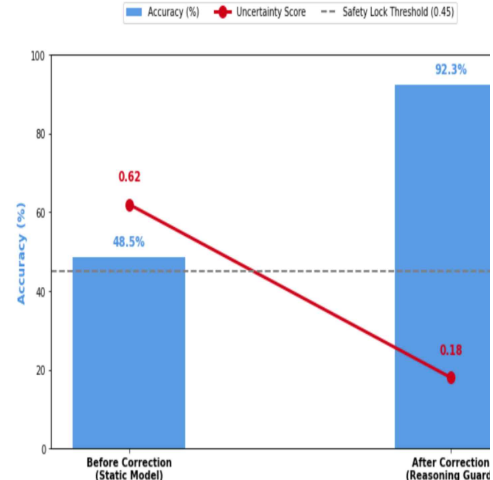


Figure 5 : Effect of AdaBoost-driven Self-Correction

The initial model (Static Model) exhibited a low accuracy of 48.5% and a high uncertainty score of 0.62 for edge cases, such as camouflaged objects or tractors in the battlefield. This indicates that the conventional model is unsuitable for making tactical decisions, falling within the Safety Lock activation zone. However, after passing through the Reasoning Guard's Self-Correction pipeline, the accuracy for the same Hard Samples significantly increased to 92.3%, while the uncertainty score decreased to 0.18. This confirms that the system successfully transitioned into the stable High-Confidence Zone. This result experimentally demonstrates that the proposed framework goes beyond merely preventing misfires; it converts its own ignorance into training data, allowing the system to actively evolve.

5. Conclusion

In this study, we addressed the vulnerability of deterministic AI in modern warfare by proposing the Reasoning Guard framework. As illustrated in the conceptual architecture of Figure 1, our approach fundamentally shifts the paradigm from simple binary classification to a probabilistic safety mechanism that acknowledges

uncertainty.

The validity of this framework was rigorously verified through our experiments. The statistical analysis in Figures 2 and 3 provided a mathematical basis for distinguishing between confidence and hesitation, while the qualitative case study in Figure 4 demonstrated that the system successfully engaged the hardware Safety Lock to prevent misfire on edge cases like tractors. These results confirm that our model possesses the "metacognitive" ability to stop firing when the risk of error is high.

Furthermore, as evidenced by the self-correction performance in Figure 5, the system does not merely stop at prevention. By utilizing AdaBoost to retrain on hard samples, we achieved a dramatic accuracy improvement (48.5% → 92.3%) and uncertainty reduction. This proves that Reasoning Guard is not just a static safety filter, but a self-evolving intelligence that turns battlefield ambiguity into a driving force for tactical reliability.

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